

$^{27}\text{Al}(^{14}\text{N},\alpha\text{pn}\gamma)$ **1976Wa11**

1976Wa11: E=40 MeV ^{14}N beam was produced at the Brookhaven National Laboratory. Target was ^{27}Al . γ rays were detected with Ge(Li) detectors. Measured E_γ , I_γ , $\gamma(\theta)$, $\gamma\gamma$ -coin, recoils distances. Deduced levels, J, π , $T_{1/2}$, γ -ray branching ratios, multipolarities.

 ^{35}Cl Levels

E(level) [†]	J π [‡]	$T_{1/2}$	Comments
0	3/2 ⁺		
1763.19 10	5/2 ⁺		
2645.45 12	7/2 ⁺		
3001.98 22	5/2 ⁺		
3162.64 9	7/2 ⁻	29.0 ps 13	$T_{1/2}$: from $\tau=41.8$ ps 18 (1976Wa11).
3942.3 10	9/2 ⁺		
4347.43 17	9/2 ⁻		
5406.68 19	11/2 ⁻		
5926.47 29	11/2 ⁻		
6086.89 23	13/2 ⁻	5.3 ps 10	$T_{1/2}$: from $\tau=7.7$ ps 14 (1976Wa11).
7872.78 30	13/2 ⁺		$T_{1/2}$: 0.5 ps< τ <50 ps (1976Wa11).
8844.2 4	17/2 ⁺	5.5 ps 14	$T_{1/2}$: from $\tau=8$ ps 2 (1976Wa11).

[†] From a least-squares fit to γ -ray energies.

[‡] As given in **1976Wa11** based on measured $\gamma(\theta)$ and known assignments of low-lying levels. When considered in Adopted Levels, firm assignments here will be placed in parentheses if there are no strong supporting arguments.

 $\gamma(^{35}\text{Cl})$

A_2 and A_4 under comments are from $\gamma(\theta)$ in **1976Wa11**, with $A_4=0$ if not given.

E_γ [†]	I_γ [†]	$E_i(\text{level})$	J π_i	E_f	J π_f	Comments
160.66 [#] 20	2.3	3162.64	7/2 ⁻	3001.98	5/2 ⁺	$A_2=-0.24$ 15.
517.26 10	<2.2 [‡]	3162.64	7/2 ⁻	2645.45	7/2 ⁺	E_γ, I_γ : doublet with γ rays in ^{36}Cl (1976Wa11). $A_2=+0.47$ 25.
680.22 15	25	6086.89	13/2 ⁻	5406.68	11/2 ⁻	$A_2=-0.20$ 2.
882.32 35	2.3	2645.45	7/2 ⁺	1763.19	5/2 ⁺	$A_2=+0.31$ 40.
971.38 20	8.0	8844.2	17/2 ⁺	7872.78	13/2 ⁺	$A_2=+0.28$ 7, $A_4=-0.18$ 7.
1059.17 20	4.6	5406.68	11/2 ⁻	4347.43	9/2 ⁻	$A_2=-0.04$ 15. %Branching=13 4.
1184.80 20	21	4347.43	9/2 ⁻	3162.64	7/2 ⁻	$A_2=-0.66$ 4, $A_4=+0.08$ 4. %Branching=69 5.
1579.15 30	1.9 [‡]	5926.47	11/2 ⁻	4347.43	9/2 ⁻	$A_2=-0.6$ 4.
1701.85 25	9.5	4347.43	9/2 ⁻	2645.45	7/2 ⁺	$A_2=-0.20$ 3. %Branching=31 5.
1739.3 4	1.7	6086.89	13/2 ⁻	4347.43	9/2 ⁻	
1763.15 10	32	1763.19	5/2 ⁺	0	3/2 ⁺	$A_2=-0.28$ 2, $A_4=+0.05$ 3.
1786 [#]	0.63	7872.78	13/2 ⁺	6086.89	13/2 ⁻	E_γ, I_γ : inferred from the presence of 680 γ -971 γ coin (1976Wa11). %Branching=8 5.
1946.40 30	1.9	7872.78	13/2 ⁺	5926.47	11/2 ⁻	$A_2=-0.2$ 1. %Branching=22 5.
2179	5.0	3942.3	9/2 ⁺	1763.19	5/2 ⁺	E_γ, I_γ : observed in $\gamma\gamma$ -coin only (1976Wa11). $A_2=+0.23$ 2, $A_4=-0.08$ 2.
2244.00 25	32	5406.68	11/2 ⁻	3162.64	7/2 ⁻	%Branching=87 4.
2465.85 30	5.9	7872.78	13/2 ⁺	5406.68	11/2 ⁻	$A_2=-0.21$ 6. %Branching=70 5.

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$^{27}\text{Al}(^{14}\text{N},\alpha\text{pn}\gamma)$ **1976Wa11** (continued) $\gamma(^{35}\text{Cl})$ (continued)

E_γ [†]	I_γ [†]	$E_i(\text{level})$	J_i^π	E_f	J_f^π	Comments
2645.79 30	<26	2645.45	7/2 ⁺	0	3/2 ⁺	E_γ, I_γ : doublet with γ rays in ^{38}K (1976Wa11). $A_2=+0.37$ 3, $A_4=-0.04$ 3.
3162.43 10	100	3162.64	7/2 ⁻	0	3/2 ⁺	

[†] From **1976Wa11**. Original values of relative intensities have been renormalized by the evaluators to $I(3162\gamma)=100$.

[‡] Estimated from $\gamma\gamma$ -coin data or known branching ratios (**1976Wa11**).

Placement of transition in the level scheme is uncertain.

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Legend

Level Scheme

Intensities: Relative I_γ

- $\longrightarrow$ $I_\gamma < 2\% \times I_\gamma^{\text{max}}$
 $\longrightarrow$ $I_\gamma < 10\% \times I_\gamma^{\text{max}}$
 $\longrightarrow$ $I_\gamma > 10\% \times I_\gamma^{\text{max}}$
 $\longrightarrow$ γ Decay (Uncertain)

